

CHRISTOPHER GARLAND

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SUMMARY

Software Development graduate with hands-on experience in systems configuration, data-center hardware builds, quality validation, and automation-supported testing. Skilled at translating technical requirements into repeatable workflows, troubleshooting hardware and configuration issues, and building practical tools with Python, Scala, and SQL.

TECHNICAL SKILLS

Languages: Python, Java, Scala, C++, JavaScript, SQL, PowerShell

Frameworks & Libraries: Streamlit, SQLAlchemy, Pytest

Tools: Git, GitHub Actions, VS Code, IntelliJ, Excel, ticketing systems

Concepts: Data structures, algorithm design, OOP, recursive descent parsing, REPL design, ETL, ORM modeling, QA validation, technical documentation

Systems: Server rack integration, hardware configuration, BIOS/UEFI & firmware updates, Dell iDRAC, Cisco switch configuration, automated testing

PROJECTS

Satisfactory Tools | *Python, Streamlit, SQLAlchemy, SQLite, Pytest* · [Live demo](#)

- Built and deployed a full-stack factory-planning web app for the game Satisfactory (live on Streamlit Community Cloud) that models production chains and tracks resource nodes, factories, and power/material balance across groups of bases.
- Wrote a pure-functional production-chain calculator that recursively walks the recipe graph — aggregating per-subtree totals (materials, byproducts, building counts, power) with cycle guarding and per-item alternate-recipe overrides.
- Designed a normalized SQLAlchemy 2.x ORM schema with relationships and cascade deletes, plus an ETL pipeline that parses the game's Docs.json export into queryable items, recipes, buildings, and factory records.
- Built a multipage Streamlit interface with confirm-gated modal CRUD, cached queries, persistent session state, JSON import/export, and domain touches like belt-tier overflow warnings and extractor auto-fill.
- Backed the app with a 29-test Pytest suite (in-memory SQLite fixtures) covering calculator math and the full CRUD layer — cascade deletes, import/export round-trips — run on every push via GitHub Actions CI.

MiniCalc — Functional Expression Interpreter | *Scala 3*

- Built a purely functional interpreter for a custom expression language — a tokenizer, a recursive-descent parser with correct operator precedence and associativity, and a tree-walking evaluator over an immutable environment, supporting arithmetic, boolean logic, let-bindings, and conditionals.
- Modeled the AST with sealed-trait algebraic data types for compiler-checked exhaustive pattern matching, handled errors functionally with Either (no exceptions) and clear type-mismatch messages, and verified behavior with a 530+ test suite spanning parsing, evaluation, error paths, and the REPL.

PROFESSIONAL EXPERIENCE

CDW — Configuration Technician, Tech Level Two

Sep 2024 – Present | Chicago-area, IL

- Hired full-time as a Tech Level Two technician after a 1+ year contract in device configuration and QC, advancing into complex enterprise full-rack builds.
- Build and integrate enterprise server racks — mounting, cabling, and labeling servers, switches, and power hardware to spec — handling 5 to 20+ devices per rack and scaling to ~4–5 racks/week with order volume.
- Configure and validate system readiness through BIOS/UEFI and firmware updates, Dell iDRAC remote-management setup, and Cisco switch firmware updates and configuration against customer-specific requirements.
- Run automated test and validation on every build, diagnosing and isolating hardware and configuration faults and re-testing before sign-off, maintaining a clean QC record to date.
- Built Python/PowerShell automation for repetitive setup steps — including a port-naming tool that replaces ~15 manual entries per unit — cutting per-build configuration time and removing a common source of manual-entry errors.
- Prepare completed racks for deployment-ready shipment, verifying asset tags, labeling, hardware security, documentation, and order accuracy for error-free customer handoff.

EDUCATION

Northern Illinois University — B.S. Software Development, GPA: 3.7

DeKalb, IL | Dec 2025

Harper College — A.A.S. Network Administration

Palatine, IL | Dec 2020

ADDITIONAL EXPERIENCE

Bonefish Grill — Server

Schaumburg, IL | Nov 2021 – Sep 2024

Bahama Breeze — Server / Bartender / Cook / Expeditor

Schaumburg, IL | Jul 2016 – Oct 2021